STAT6611

- Lecture 07 (T, Sept 15)
- Homework 1 assigned
  - practice on g.i.2 matrix results
  - due F, Sept 18 (F next week) by midnight in gradescope/canvas
  - My office hours: TW 12:30-1:30 PM & R 1-2 PM
    TA Ben: R 9:30-10:30 AM in Snedecor 2207
- Homework 2 will be assigned this Thursday (again due Friday in 2 weeks)

Definitions. If $\underline{x}$ is a column vector of r real variables and A is an $r \times r$ matrix of constants, then we refer to the function

$$q(\underline{x}) = \underline{x}' A \underline{x}$$

as a quadratic form, where quadratic form $q(\underline{x})$ is said to be

nonnegative definite if $q(\underline{x}) \geq 0$ for all real vectors $\underline{x}$;

positive definite if $q(\underline{x}) > 0$ for all nonzero real vectors $\underline{x}$;

positive semidefinite if it is nonnegative definite but not positive definite (i.e., $q(\underline{x}) \geq 0$ with equality holding for at least one nonzero real vector $\underline{x}$)

A symmetric square matrix A is nonnegative definite, positive definite, positive semidefinite according to whether quadratic form $\underline{x}' A \underline{x}$ is nonnegative definite, positive definite, or positive semidefinite, respectively.

Theorem 2.10. *Under Model II, suppose that $\Lambda'\underline{\beta}$ is any k-dimensional estimable vector and that $\underline{c} + A'\underline{y}$ is any vector of linear unbiased estimators of the elements of $\Lambda'\underline{\beta}$. Then, the matrix $\mathrm{Var}(\underline{c} + A'\underline{y}) - \mathrm{Var}(\Lambda'\widehat{\underline{\beta}})$ is nonnegative definite, where $\widehat{\underline{\beta}}$ is any solution to the normal equations.*

Proof. For any $k \times 1$ vector $\underline{\ell}$, by the corollary to Lemma 2.4,

$$\underline{\ell}'[\mathrm{Var}(\underline{c}+A'\underline{y}) - \mathrm{Var}(\Lambda'\widehat{\underline{\beta}})]\underline{\ell} = \underline{\ell}'\mathrm{Var}(\underline{c}+A'\underline{y})\underline{\ell} - \underline{\ell}'\mathrm{Var}(\Lambda'\widehat{\underline{\beta}})\underline{\ell} = \mathrm{Var}[\underline{\ell}'(\underline{c}+A'\underline{y})] - \mathrm{Var}(\underline{\ell}'\Lambda'\widehat{\underline{\beta}}).$$

By Cor 2/Theorem 2.1, $\tilde{\underline{\lambda}}'\underline{\beta}$ for $\tilde{\underline{\lambda}}' \equiv \underline{\ell}'\Lambda'$ is estimable with least squares estimator as $\tilde{\underline{\lambda}}'\widehat{\underline{\beta}}$. Since $\underline{\ell}'(\underline{c} + A'\underline{y})$ is a linear unbiased estimator of $\tilde{\underline{\lambda}}'\underline{\beta} = (\underline{\ell}'\Lambda')\underline{\beta}$,

Corollary 1. Under the condition of Theorem 2.10,

$$\mathrm{tr}[\mathrm{Var}(\underline{c} + A'\underline{y})] \geq \mathrm{tr}[\mathrm{Var}(\Lambda'\underline{\hat{\beta}})].$$

Let $\underline{\ell}_i$ be the ith coordinate vector in $\mathbb{R}^p$. Since $U - W$ is n.n.d,

$$0 \leq \underline{\ell}_i'(U - W)\underline{\ell}_i = \underline{\ell}_i' U \underline{\ell}_i - \underline{\ell}_i' W \underline{\ell}_i \leftarrow \text{this } i\text{th diagonal element of } W$$

$$\Rightarrow \mathrm{tr}(U) = \sum_{i=1}^{p} \underline{\ell}_i' U \underline{\ell}_i \geq \sum_{i=1}^{p} \underline{\ell}_i' W \underline{\ell}_i = \mathrm{tr}(W) \quad \leftarrow \text{this is } i\text{th diagonal of } U$$

Corollary 2. Under the condition of Theorem 2.10,

$$\det[\mathrm{Var}(\underline{c} + A'\underline{y})] \geq \det[\mathrm{Var}(\Lambda'\underline{\hat{\beta}})]. \quad \leftarrow \Lambda'\hat{\beta} \text{ has smallest generalized variance}$$

Definition. The determinant of the variance-covariance matrix of an estimator is referred to as the generalized variance of the estimator.

Remark. (This material is optional.) To prove Corollary 2, we use Theorem 12.2.14 (7) in the first edition of Graybill's book *Introduction to Matrices with Applications in Statistics*, as stated in Result 1 below, which extends to Result 2 below. Corollary 2 follows from Result 2 with

$$U = \mathrm{Var}(\underline{c} + A'\underline{y}), \quad V = \mathrm{Var}(\underline{c} + A'\underline{y}) - \mathrm{Var}(\Lambda'\underline{\hat{\beta}}), \quad \text{and} \quad U - V = \mathrm{Var}(\Lambda'\underline{\hat{\beta}}).$$

n.n.d; n.n.d by Theorem 2.10; n.n.d

Result 1. For $m \times m$ matrices U and V, if U is positive definite and V and $U - V$ are nonnegative definite, then $\det(U) \geq \det(U - V)$.

Result 2. For $m \times m$ matrices U and V, if U, V, and $U - V$ are nonnegative definite, then $\det(U) \geq \det(U - V)$.

Proof. If U is positive definite, Result 2 becomes Result 1. If U is positive semidefinite, then there exits a nonzero vector $\underline{t}$ such that $\underline{t}'U\underline{t} = 0$. Thus, $0 = \underline{t}'U\underline{t} = \underline{t}'[V + (U - V)]\underline{t} = \underline{t}'V\underline{t} + \underline{t}'(U - V)\underline{t}$, implying $\underline{t}'(U - V)\underline{t} = 0$ as V and $U - V$ are nonnegative definite. Then, U and $U - V$ are positive semidefinite (not full rank) and $\det(U) = 0 = \det(U - V)$ hold. □

2.8 Estimation of the Residual Variance

We now consider the estimation of σ^2 for Model II. We begin by introducing some lemmas that will be useful in this and subsequent discussions.

Definition. A square matrix A is said to be an idempotent matrix if $A^2 = A$.

Lemma 2.5. *For any square matrix A such that $A^2 = mA$ for some scalar m,*

$$\mathrm{tr}(A) = m\mathrm{rank}(A).$$

Proof. This holds when $A = 0$. For $A \neq 0$, $r = \mathrm{rank}(A) \geq 1$. By the proof of Theorem 1.1, there exist nonsingular matrices P and Q such that

$$PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix},$$

where I_r is the $r \times r$ identity matrix. We have

$$A^2 = mA \iff \left(P^{-1}\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}Q^{-1}\right)\left(P^{-1}\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}Q^{-1}\right) = m\left(P^{-1}\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}Q^{-1}\right)$$

$$\iff \left(\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}Q^{-1}\right)\left(P^{-1}\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}\right) = m\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}.$$

Note that for any $p \times q$ matrix F and $q \times p$ matrix G, $\mathrm{tr}(FG) = \mathrm{tr}(GF)$.

Thus,

$$m\mathrm{rank}(A) = \mathrm{tr}\left(m\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}\right) = \mathrm{tr}\left\{\left(\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}Q^{-1}\right)\left(P^{-1}\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}\right)\right\}$$

$$= \mathrm{tr}\left\{\left(P^{-1}\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}\right)\left(\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}Q^{-1}\right)\right\} = \mathrm{tr}\left(P^{-1}\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}Q^{-1}\right) = \mathrm{tr}(A).$$

□

Corollary. If A is idempotent, then $\mathrm{tr}(A) = \mathrm{rank}(A)$.

Recall the following properties of the projection matrix $P_X = XB$ where B is a solution to $X'XB = X'$ (after Lemma 1.5):

(1) P_X is invariant to the choice of solution B,

(2) P_X is symmetric,

(3) $P_X = X(X'X)^{-}X'$,

(4) $P_X X = X$,

(5) P_X is idempotent, that is, $P_X^2 = P_X$.

Here X is $n \times p$ so I & P_X are $n \times n$

It follows that $(I - P_X)$ is also symmetric and idempotent.

$(I-P_X)' = I' - P_X' = I - P_X$

$(I-P_X)^2 = I - 2P_X + P_X^2 = I - P_X$

$\sim P_X$

Lemma 2.6. *For any matrix A, P_A and $I - P_A$ are both symmetric idempotent matrices, where $P_A = A(A'A)^{-}A'$.*

$= tr(P_A)$

Lemma 2.7. *For any $n \times p$ matrix A,* $\text{rank}(P_A) = \text{rank}(A)$ *and* $\text{rank}(I - P_A) = n - \text{rank}(A)$.

$tr(I-P_A)$
$= tr(I) - tr(P_A)$
$= n - \text{rank}\, A$

Note: $\text{rank}(A) = \text{rank}(P_A A)$
$\leq \text{rank}(P_A) = \text{rank}(A(A'A)^{-}A')$
$\leq \text{rank}(A)$ //

Lemma 2.8. *If $\underline{x}$ is an $n \times 1$ random vector having mean vector $\underline{\mu}$ and variance-covariance matrix Σ and if A is an $n \times n$ matrix of constants, then*

$$E(\underline{x}'A\underline{x}) = \text{tr}(A\Sigma) + \underline{\mu}'A\underline{\mu}.$$

$E(\underline{x}) = \mu$
$Var(\underline{x}) = \Sigma$

Proof.

$$E(\underline{x}'A\underline{x}) = E[\text{tr}(\underline{x}'A\underline{x})] = E[\text{tr}(A\underline{x}\,\underline{x}')] = \text{tr}[E(A\underline{x}\,\underline{x}')] = \text{tr}[AE(\underline{x}\,\underline{x}')]$$

$$= \text{tr}[A(\Sigma + \underline{\mu}\,\underline{\mu}')] = \text{tr}(A\Sigma) + \text{tr}(A\underline{\mu}\,\underline{\mu}') = \text{tr}(A\Sigma) + \text{tr}(\underline{\mu}'A\underline{\mu}) = \text{tr}(A\Sigma) + \underline{\mu}'A\underline{\mu}.$$

□

Let $\widehat{\underline{\beta}}$ be any solution to the normal equationsand take the sum of squares of the deviations of the elements of $\underline{y}$ from the elements of $X\widehat{\underline{\beta}}$ as

$$Q(\widehat{\underline{\beta}}) = (\underline{y}-X\widehat{\underline{\beta}})'(\underline{y}-X\widehat{\underline{\beta}}) = (\underline{y}-P_X\underline{y})'(\underline{y}-P_X\underline{y}) = [(I-P_X)\underline{y}]'[(I-P_X)\underline{y}] = \underline{y}'(I-P_X)\underline{y}.$$

↑ sum of squared residuals $\hat{\underline{e}} = \underline{Y} - \hat{\underline{Y}} = \underline{Y} - X\hat{\underline{\beta}}$ $\quad X\hat{\underline{\beta}} = P_X\underline{Y}$

Under Model II, we have $E[Q(\widehat{\underline{\beta}})] =$ $E(\underline{Y}'A\underline{Y})$ for $A = I - P_X$

$\underline{\mu} = E(\underline{Y}) = X\underline{\beta}$

$= tr(A\Sigma) + \underline{\mu}'A\underline{\mu}$ & $Var(\underline{Y}) = \Sigma$

$= \sigma^2 I_n$

$= tr((I-P_X)\sigma^2 \cdot I_n)$

$+ (X\underline{\beta})'(I-P_X)X\underline{\beta}$

$= 0$ since $(I-P_X)X = X - X$

$= 0$

$= tr(I-P_X)\sigma^2$

$= \sigma^2(tr(I) - tr(P_X))$

$= \sigma^2(n - rank(X))$

$= \sigma^2(n-p^*)$, $p^* = rank(X)$

Thus, under Model II,

$$\widehat{\sigma}^2 = (\underline{y} - X\widehat{\underline{\beta}})'(\underline{y} - X\widehat{\underline{\beta}})/(n-p^*) = \underline{y}'(I-P_X)\underline{y}/(n-p^*) = \underline{y}'(\underline{y} - X\widehat{\underline{\beta}})/(n-p^*)$$

is an unbiased estimator of σ^2.

MSE i.e. $E(\hat{\sigma}^2) = E\left(Q(\hat{\underline{\beta}})/n-p^*\right) = \sigma^2$

2.9 Constrained Linear Models

In Models I—VII, $\underline{\beta}$ can assume any value in the Euclidean p-space. We now consider the case where $\underline{\beta}$ is restricted to the subspace of R^p consisting of $\underline{\beta}$ values that satisfy

$$A\underline{\beta} = \underline{b},$$

where A is a given $q \times p$ matrix of rank $q^* \geq 1$ and $\underline{b}$ is a given $q \times 1$ vector. To avoid trivialities, we assume that the linear system $A\underline{\beta} = \underline{b}$ is consistent. We will refer to those modified versions of Models I—VII in which $\underline{\beta}$ is restricted to this subspace as Models Ia—VIIa.

Definition. Under Models IIa—VIIa, an estimator $t(\underline{y})$ is said to be an unbiased estimator of a linear function $\underline{\lambda}'\underline{\beta}$ if $E[t(\underline{y})] = \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.

Lemma 2.9. *Under Models IIa—VIIa, a linear estimator $c + \underline{a}'\underline{y}$ of a linear function $\underline{\lambda}'\underline{\beta}$ is unbiased if and only if there exists a vector $\underline{k}$ such that*

$$c = \underline{k}'\underline{b} \quad \text{and} \quad \underline{\lambda}' = \underline{a}'X + \underline{k}'A.$$